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Fintech, Regulatory Arbitrage, and the Rise of Shadow Banks

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1 Big picture

- **Question.** Why did nonbank mortgage originators — especially shadow banks and online “fintech” lenders — gain so much market share after the financial crisis?
- **Two candidate forces.** The paper studies two broad explanations. First, *regulatory arbitrage*: post-crisis regulation made mortgage lending relatively harder for depository banks, while non-depository lenders were not subject to the same constraints. Second, *technology*: online lenders improved convenience and perhaps underwriting, allowing them to win borrowers even without cheaper rates.
- **Main fact.** Shadow banks’ share of residential mortgage originations rises from roughly 30% in 2007 to about 50% in 2015. In FHA lending, their share rises from about 45% to about 75%. Fintech lenders rise from about 3% of all mortgages in 2007 to about 12% in 2015.
- **Where shadow banks grow.** Shadow banks are especially active among FHA borrowers, lower-income borrowers, minority-heavy counties, and areas where local banks are more exposed to post-crisis regulation.
- **Where fintech grows.** Fintech lenders are especially active in refinancing and among borrowers who appear more willing to pay for convenience. They do not generally offer cheaper conforming mortgages; instead, they charge a noticeable premium.
- **Main reduced-form conclusion.** Counties whose banks faced bigger increases in regulatory burden — capital rebuilding, mortgage servicing rights (MSR) treatment, mortgage-related lawsuits, and tighter supervision after the OTS closure — experienced larger gains in shadow-bank market share and lending.
- **Main pricing conclusion.** Relative to other shadow banks, fintech lenders charge about 14–16 basis points more on conforming mortgages, suggesting that technology is not mainly a cost-cutting force passed through as lower prices. It looks more like a quality or convenience improvement.
- **Main technology conclusion.** Observable “hard” information such as FICO and LTV explains less of fintech pricing than non-fintech pricing, consistent with fintech lenders using different data or scoring models.
- **Quantitative decomposition.** The paper’s calibrated mortgage-market model attributes roughly 60% of shadow-bank growth from 2008 to 2015 to regulatory changes that impaired banks, and roughly 30% to technological improvements associated with fintech. Technology explains about 90% of fintech’s own growth.
- **Economic takeaway.** The post-crisis mortgage market did not merely move from “banks” to “cheaper lenders.” It moved toward institutions that were less regulated on the origination side and, in fintech’s case, better able to sell convenience and new screening technology, while still relying heavily on public credit guarantees through the GSEs and FHA.

2 Data and measurement (key objects)

2.1 Data sources

The paper combines several large data sets.

- **HMDA, 2007–2015.** This is the broad market-wide origination data set. It records lender identity, borrower income, race, loan purpose, loan type, and whether the loan is sold within the calendar year.
- **Fannie Mae and Freddie Mac loan-level data.** These cover a large share of conforming fixed-rate mortgages and include richer borrower and contract information such as interest rates, FICO scores, LTV ratios, debt-to-income ratios, and subsequent loan performance.
- **FHA servicing snapshots, 2010–2016.** These are used to study FHA lending, an especially important market for shadow banks.
- **County demographics from the Census and ACS.** These provide income, race, unemployment, education, and homeownership measures for the county-level analysis.
- **Bank call reports and supervisory data.** These are used to measure bank capital ratios and regulatory exposure.
- **Mortgage-lawuit data.** The paper assembles mortgage-related settlements from Law360, SEC actions, and SNL to proxy for legal and regulatory pressure borne mainly by banks.

Interpretation. The data structure is designed to answer three distinct questions: who gained market share, how those lenders differ at the loan level, and whether that growth lines up with local exposure to bank-specific regulation.

2.2 Lender classification

A lender is classified as a **bank** if it is a depository institution and as a **shadow bank** otherwise. Within shadow banks, a lender is classified as **fintech** if nearly all of the application and underwriting process takes place online with very limited human interaction before closing.

The authors do this classification manually and verify it using lender websites and archived versions of websites. Their classified sample covers about 80% of total origination value in 2010.

Interpretation. “Shadow bank” here means nondepository mortgage originator. “Fintech” is not just “has a website”; it means online-first origination with minimal loan-officer intermediation.

2.3 Market-share objects

The broad market-share variables are:

- shadow-bank share of all mortgage originations,
- shadow-bank share within conforming loans,
- shadow-bank share within FHA loans,
- fintech share of all originations,
- and fintech share within shadow-bank originations.

Key descriptive facts are:

- shadow-bank share rises from roughly 30% to 50% between 2007 and 2015,
- conforming shadow-bank share rises from roughly 25% to 50%,
- FHA shadow-bank share rises from roughly 45% to 75%,
- fintech share of all mortgages rises from roughly 3% to 12%.

Interpretation. The rise is not confined to one niche. It is visible in the overall mortgage market, in GSE-eligible conforming loans, and most dramatically in FHA lending.

2.4 Financing and disposition measures

HMDA records whether a loan is held or sold within the year and, if sold, to whom. This allows the paper to compare banks and shadow banks on funding structure.

Between 2007 and 2015:

- traditional banks keep more than 30% of originated mortgages on balance sheet,
- shadow banks retain at most about 7.5%,
- shadow banks become increasingly reliant on GSE execution,
- fintech lenders also shift heavily toward GSE-backed funding after 2010.

Interpretation. Traditional banks use deposits and balance-sheet capacity. Shadow banks use originate-to-distribute. This difference becomes even sharper after the crisis, when private securitization collapses and GSE/FHA guarantees become central.

2.5 Time to sale

For GSE-eligible conforming loans, the paper measures how quickly an originated loan is sold after origination.

The average loan is sold in about 0.46 quarters, roughly 41 days. Among shadow-bank loans, the mean is about 0.40 quarters, roughly 36 days.

Interpretation. Faster sale is consistent with a more pure originate-to-distribute model and less reliance on warehouse or portfolio balance-sheet capacity.

2.6 Borrower and loan characteristics

The paper studies borrower and loan characteristics in two ways:

- **individual-level within-market comparisons** using county-time or zip-quarter fixed effects,
- **county-level penetration regressions** using demographic and market-structure controls.

Main variables include borrower income, race, sex, loan purpose (purchase, refinance, home improvement), loan type (conventional, FHA, VA, FSA/RHS), FICO, LTV, debt-to-income ratio, first-time buyer status, and county characteristics such as unemployment and racial composition.

Interpretation. This is how the paper connects regulation and technology to actual borrower segments instead of only aggregate market shares.

2.7 Loan pricing and performance

For conforming loans in the Fannie/Freddie data, the authors observe:

- the mortgage interest rate,
- ex-post default, defined as becoming 60 days delinquent within two years,
- ex-post prepayment within two years.

Interpretation. Rates tell us whether shadow banks and fintech win by charging less. Performance tells us whether their borrower pools are observationally similar and whether investors face different risks.

2.8 Regulatory exposure measures

At the county level, the paper constructs four measures of how exposed local banks were to post-crisis regulatory pressure:

- increases in bank tier-1 capital ratios from 2008 to 2015,
- pre-existing MSR intensity as a share of tier-1 capital,
- exposure to mortgage-related lawsuits,
- and the 2008 share of originations made by OTS-regulated institutions.

Each measure is aggregated to the county using banks' 2008 origination shares as weights.

Interpretation. Counties are treated as more “exposed” if the banks active there before the post-crisis reforms were exactly the banks hit hardest by those reforms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Originate-to-distribute timing

The first timing object is:

$$\text{TimeToSale}_{ijzt} = \text{QuartersBetween}(\text{Sale}, \text{Origination}). \quad (1)$$

The regression is:

$$\text{TimeToSale}_{ijzt} = \beta \text{Type}_j + X_i' \Gamma + \delta_{zt} + \varepsilon_{ijzt}. \quad (2)$$

Here, i is the loan, j the lender type, z the zip code, and t the quarter. The controls X_i include borrower and loan characteristics, while δ_{zt} are zip-quarter fixed effects.

Interpretation. This regression asks: for two otherwise similar loans originated in the same local market at the same time, does the lender type predict how quickly the loan is sold? A shorter time-to-sale indicates a more purely distributive funding model.

3.2 Loan-level lender-type regressions

The paper’s main within-market composition regression is:

$$\text{LenderType}_{ict} = X'_i \Gamma + \delta_{ct} + \varepsilon_{ict}. \quad (3)$$

The left-hand side is a linear-probability indicator. In one version it equals 100 if the lender is a shadow bank; in another it equals 100 if, among shadow-bank loans, the lender is fintech.

Interpretation. This is the core descriptive design for identifying which borrowers and loan types sort into shadow banks and fintech once one compares loans in the same county and time period. It is not causal by itself, but it is very useful for mapping lender specialization.

3.3 County-level penetration regressions

To study cross-county heterogeneity, the paper estimates:

$$\% \text{LenderTypeLoans}_c = X'_c \Gamma + \varepsilon_c. \quad (4)$$

The two main market-share objects are:

$$\% \text{ShadowBankLoans}_c = \frac{\sum_{i \in \text{shadow}} \text{DollarsOriginated}_{ic}}{\sum_{i \in \text{all}} \text{DollarsOriginated}_{ic}}, \quad (5)$$

$$\% \text{FintechLoans}_c = \frac{\sum_{i \in \text{fintech}} \text{DollarsOriginated}_{ic}}{\sum_{i \in \text{shadow}} \text{DollarsOriginated}_{ic}}. \quad (6)$$

Interpretation. These county regressions complement the within-market borrower regressions. They ask whether counties with more minorities, more unemployment, more refinancing potential, or different market structure also have systematically higher shadow-bank or fintech penetration.

3.4 Mortgage-rate regression

For conforming loans, the pricing specification is:

$$\text{Rate}_{ijzt} = \beta_1 \text{NonFintechSB}_j + \beta_2 \text{FintechSB}_j + X'_i \Gamma + \delta_{zt} + \varepsilon_{ijzt}. \quad (7)$$

The authors compare rates within the same zip-quarter and control for loan and borrower observables such as FICO, LTV, and debt-to-income.

Interpretation. This is a strong apples-to-apples pricing comparison. If shadow banks or fintech win mainly by being cheaper, the coefficients should be negative. If fintech instead sells convenience or differentiated quality, its coefficient can be positive.

3.5 Default and prepayment regressions

For conforming loans with two years of subsequent performance, the paper estimates:

$$\text{Default}_{ijzt} = \beta_1 \text{NonFintechSB}_j + \beta_2 \text{FintechSB}_j + \beta_r \text{Rate}_{ijzt} + X_i' \Gamma + \delta_{zt} + \varepsilon_{ijzt}, \quad (8)$$

$$\text{Prepayment}_{ijzt} = \beta_1 \text{NonFintechSB}_j + \beta_2 \text{FintechSB}_j + \beta_r \text{Rate}_{ijzt} + X_i' \Gamma + \delta_{zt} + \varepsilon_{ijzt}. \quad (9)$$

Default means becoming at least 60 days delinquent within two years; prepayment is defined analogously.

Interpretation. These regressions ask whether investors end up bearing different ex-post risks from loans originated by banks, non-fintech shadow banks, and fintech lenders, conditional on observables and the loan's interest rate.

3.6 Regulatory-exposure construction

The bank-capital shock starts with bank-level capital rebuilding:

$$\Delta CR_b = \text{Tier1RBC}_{b,2015} - \text{Tier1RBC}_{b,2008}. \quad (10)$$

This is aggregated to the county using 2008 origination shares:

$$\Delta \text{LocalCapitalRatio}_c = 100 \times \sum_{b \in c} \Delta CR_b \cdot \omega_{bc,2008}, \quad (11)$$

where $\omega_{bc,2008}$ is bank b 's share of 2008 originations in county c .

Analogously, the authors build county exposure measures for MSR intensity, lawsuit exposure, and OTS-regulated lending.

Interpretation. The weighting is crucial. It pins county exposure to the banks that were active there before the regulatory shocks became fully relevant, which reduces the danger of simply reading off post-shock market shares.

3.7 County-level regulation regressions

The two main outcomes are the change in county shadow-bank market share and the growth in county shadow-bank lending:

$$\Delta \text{ShadowBankShare}_c = 100 \left(\frac{\text{ShadowOriginations}_{c,2015}}{\text{AllOriginations}_{c,2015}} - \frac{\text{ShadowOriginations}_{c,2008}}{\text{AllOriginations}_{c,2008}} \right), \quad (12)$$

$$\Delta \text{ShadowBankLending}_c = 100 \left(\frac{\text{ShadowOriginations}_{c,2015} - \text{ShadowOriginations}_{c,2008}}{\text{AllOriginations}_{c,2008}} \right). \quad (13)$$

The estimating equations are:

$$\Delta \text{ShadowBankShare}_c = \beta_0 + \beta_1 \text{Regulation}_c + X_c' \Gamma + \varepsilon_c, \quad (14)$$

$$\Delta \text{ShadowBankLending}_c = \beta_0 + \beta_1 \text{Regulation}_c + X_c' \Gamma + \varepsilon_c. \quad (15)$$

The regulation variable is one of the four standardized county-level exposure measures.

Interpretation. The first equation asks whether shadow banks gain market share where local banks are more constrained. The second asks whether shadow banks merely gain share because banks shrink, or whether shadow banks actively expand lending into those counties.

3.8 Event-study timing for MSR and OTS shocks

For the two shocks with clear dates, the paper estimates:

$$\text{ShadowBankShare}_{ct} = \gamma_t + \gamma_c + \beta (\text{HighReg}_c \times \text{Year}_t) + (X_c \times \text{Year}_t)' \Gamma + \varepsilon_{ct}. \quad (16)$$

Interpretation. This is a timing test. If the story is really about the MSR rule and the OTS closure, the gap between high-exposure and low-exposure counties should widen right around 2013 for MSR and around 2011 for OTS.

3.9 Technology and pricing-model regressions

To ask whether fintech uses different information, the paper studies how much of the cross-loan variation in rates is explained by standard observables:

$$\text{Rate}_{izt} = \beta_1 \text{FICO}_i + \beta_2 \text{LTV}_i + X_i' \Gamma + \delta_{zt} + \varepsilon_{izt}. \quad (17)$$

The regression is run separately for banks, shadow banks, non-fintech shadow banks, and fintech shadow banks, and the object of interest is the within-fixed-effect R^2 .

Interpretation. If fintech lenders rely on different signals or more flexible models, standard hard information should explain less of their pricing variation.

3.10 Fintech convenience premium

To test whether high-value borrowers pay more for fintech convenience, the paper estimates:

$$\text{Rate}_{izt} = \beta_s \text{Fintech}_{izt} + \beta_{hs} (\text{Fintech}_{izt} \times \text{HighFICO}_{izt}) + X_i' \Gamma + \delta_{zt} + \varepsilon_{izt}. \quad (18)$$

Interpretation. The fintech coefficient measures the fintech premium for the lower 90% of the FICO distribution. The interaction shows whether the highest-FICO borrowers — likely less price-sensitive and more convenience-sensitive — pay an even larger premium for fintech origination.

3.11 Quantitative mortgage-market model

The paper then introduces a simple three-lender-type discrete-choice model with banks, non-fintech shadow banks, and fintech shadow banks.

Borrower utility is:

$$u_{ib} = -\alpha r_i + q_i + \varepsilon_{ib}. \quad (19)$$

Lenders differ in service quality q_i , funding cost ρ_i , and fixed cost c_i . Banks also face a regulatory burden parameter γ_b that scales down their effective ability to lend.

Per-loan profit is:

$$(r_i - \rho_i)\gamma_i s_i, \quad (20)$$

so total profit is

$$\pi_i = (r_i - \rho_i)\gamma_i s_i F - c_i. \quad (21)$$

With logit demand, lender-type shares depend on rates, quality, and the regulatory burden. The optimal markup takes the standard form:

$$r_i^* - \rho_i = \frac{1}{\alpha} \frac{1}{1 - s_i}. \quad (22)$$

The authors then invert the observed shares and rates to calibrate:

- borrower price sensitivity α_t ,
- shadow-bank funding costs relative to banks,
- fintech quality relative to non-fintech shadow banks,
- bank regulatory burden γ_{bt} ,
- and fixed costs.

Interpretation. The model is not a full structural mortgage model. It is a disciplined decomposition tool. It translates the observed rates and shares into a small set of interpretable wedges: regulation, quality/convenience, and costs.

4 Identification and instruments (what makes the estimates credible)

4.1 Why bank versus shadow-bank comparisons speak to regulation

The central regulation argument is not just that shadow banks grew. It is that they grew most where post-crisis rules constrained banks the most. The county-level exposure measures are all constructed from *pre-existing* bank presence in 2008. That helps tie later growth in shadow banking to the local legacy footprint of regulated banks rather than to purely contemporaneous demand shifts.

Interpretation. The design is closer to “counties where banks were hit harder saw more nonbank substitution” than to a simple cross-sectional comparison of shadow-bank-heavy and shadow-bank-light places.

4.2 Why fintech versus non-fintech comparisons speak to technology

A major strength of the paper is that it compares fintech to other shadow banks. Both groups are outside depository-bank regulation, so differences between them are less likely to be driven by regulation and more likely to reflect technology, product design, data use, or convenience.

Interpretation. This is the cleanest part of the identification strategy. The bank/shadow-bank comparison is about regulation; the fintech/non-fintech comparison is about technology within a common regulatory environment.

4.3 Role of fixed effects and within-market comparisons

Many loan-level regressions use county-time or zip-quarter fixed effects. This means the comparisons are made within the same place and time, holding local credit demand and macro conditions much more tightly fixed.

Interpretation. When the paper says fintech rates are higher, it is not comparing San Francisco in 2015 to rural Ohio in 2010. It is comparing observationally similar borrowers in the same local market at the same time.

4.4 Difference-in-differences logic in the regulation tests

The four regulation measures all exploit heterogeneity in local exposure:

- counties where banks needed to rebuild more capital,
- counties where banks had more MSRs relative to capital,
- counties where local banks were more exposed to mortgage lawsuits,
- counties where OTS-supervised lenders were more important before the OTS disappeared.

The event-study graphs for MSR and OTS provide an additional timing test: the differential rise in shadow-bank share occurs when the relevant rule changes actually arrive.

Interpretation. This is not randomized policy variation, but it is much stronger than a raw time-series story. The timing lines up with the proposed shocks.

4.5 Instrument for capital rebuilding

To address endogeneity in the post-crisis increase in capital ratios, the paper instruments 2008–2015 capital-ratio increases with 2006–2008 capital-ratio declines.

The idea is simple: banks that lost more capital during the crisis had to rebuild more afterward. If counties exposed to those banks later saw more shadow-bank growth, that is evidence that bank recapitalization pressure helped displace banks from mortgage origination.

Interpretation. This IV does not make the entire regulation story bulletproof, but it strengthens the capital-channel part of the argument by linking post-crisis capital rebuilding to pre-crisis balance-sheet damage.

4.6 What remains descriptive rather than causal

Some important parts of the paper are more suggestive than fully causal:

- the fintech R^2 exercise shows different pricing information, but does not directly reveal what alternative data are used;

- the convenience interpretation of fintech’s rate premium is plausible and well supported, but not uniquely identified;
- the quantitative model is a calibrated decomposition, not an experimental estimate.

Interpretation. The paper is strongest on the claim that post-crisis regulation shifted origination away from banks and that fintech sells a differentiated product. It is somewhat less definitive on the exact microdata mechanism behind fintech underwriting.

5 Tables and figures

5.1 Figures 1–4 and Table 1: shadow banking rises and funding structures diverge

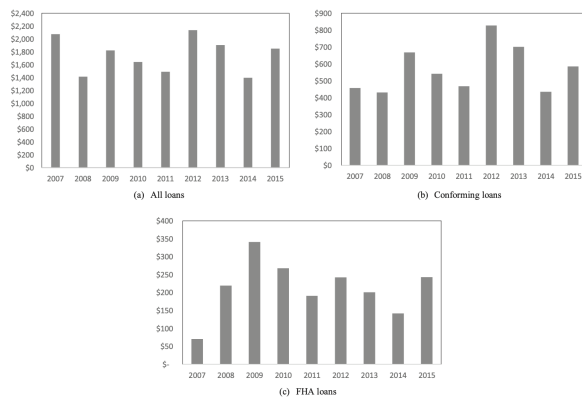


Fig. 1. Total residential mortgage originations. Panel A shows total dollars in billions originated between 2007 and 2015 as reported by HMDA. Panel B shows the total dollar value of originated conforming mortgages, where a mortgage is conforming if it is conventional and reported as sold to Fannie Mae or Freddie Mac in HMDA. Note that if the mortgage is sold to Fannie Mae or Freddie Mac more than a year after origination it is not reported as sold and hence not counted in Panel B. Panel C shows total dollars of FHA originations.

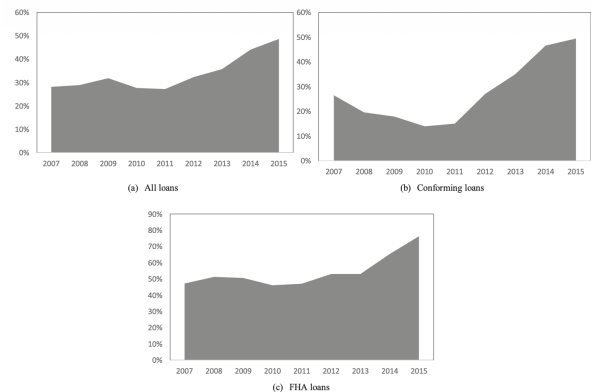


Fig. 2. Shadow bank origination shares. Panel A shows shadow bank origination shares as a fraction of total originations for all mortgages in HMDA between 2007 and 2015. Panel B shows shadow bank origination shares among conforming mortgages. Panel C shows the shadow bank origination share among FHA mortgages.

Read: Figures 1–3 show the core aggregate facts. Shadow-bank market share rises from about 30% to 50% overall, from about 25% to 50% in conforming loans, and from about 45% to 75% in FHA loans. Fintech grows from about 3% to 12% of all mortgage originations. Table 1 shows that banks hold roughly 31% of mortgages on balance sheet, versus only about 7.5% for shadow banks. It also shows that shadow banks originate far more FHA loans than banks.

Economic message. The paper starts by documenting a genuine reorganization of mortgage intermediation. This is not just a small niche moving online. It is a large shift in who originates mortgages and how those originations are funded.

5.2 Table 2: shadow banks move loans off balance sheet faster

Read: The time-to-sale regression shows that shadow banks sell conforming loans about 0.10 quarters faster than banks, roughly 9 days. Fintech shadow banks sell even faster than non-fintech shadow banks, by about 0.084 quarters. **Economic message.** Faster sale is exactly what one would expect from institutions built around originate-to-distribute. It also suggests that fintech is not just a marketing label; it is associated with operational differences in how quickly mortgages move through the pipeline.

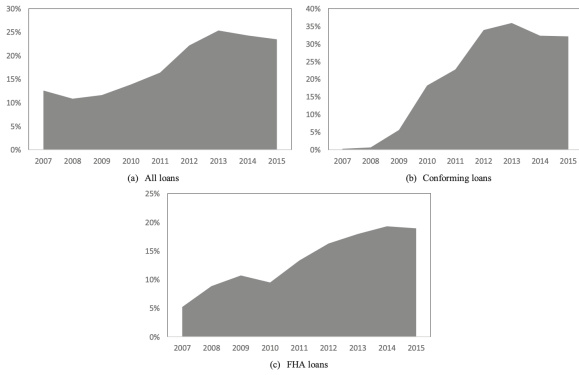


Fig. 3. Fintech origination shares of shadow bank originations. Panel A of this figure shows fintech originations as a share of shadow bank originations for all mortgages in HMDA between 2007 and 2015. Panel B shows fintech bank origination shares among shadow bank conforming originations. Panel C shows fintech share among shadow bank FHA originations (based on HMDA).

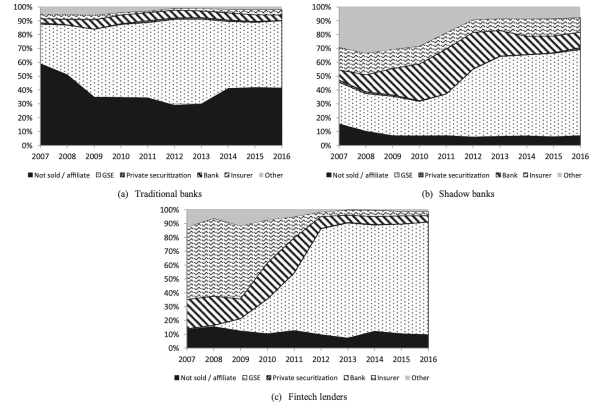


Fig. 4. Disposition of loans among traditional banks, shadow banks, and fintech lenders. Fig. 4 shows the percentage of originated loans by originator type sold to various entities within the calendar year of origination (including loans not sold). Panel A shows the buyer composition of traditional bank originations; Panel B shows the buyer composition of all shadow bank originations; Panel C shows the buyer composition of fintech shadow bank originations. Loans categorized as “unsold” are not sold within the calendar year of origination, although they may be sold some time later. The CSE category pools Fannie Mae, Freddie Mac, Ginnie Mae, and Farmer Mac. Calculations are based on HMDA data.

Table 1
Residential mortgage lending: traditional versus shadow banks.

Table 1 provides shows loan type and disposition by lender type. Panel A reports the types of loans types made by different lenders between 2007 and 2015. Loan types are Conventional, FHA, or Other, which includes VA and FSA/RHS (Farm Service Agency and Rural Housing Service) loans. Conventional loans are all loans that are not FHA or VA/FSA/RHS loans. Column (1) reports the composition of loans made by all lenders; Column (2) reports those made by traditional banks; Column (3) reports those made by shadow banks. Column (4) reports those made by non-fintech shadow banks, and Column (5) reports those made by fintech shadow banks. Panel B reports to which type of entity the originating entity sold the loan. Loans not sold within one year are Not sold. Columns are the same as in Panel A.

Panel A: loan types based on 2007–2015 HMDA					
	All lenders	Traditional banks	Shadow banks	Shadow banks	
				Non-fintech	Fintech
% Conventional	76.9%	83.2%	64.31%	62.0%	74.3%
% FHA	15.8%	11.0%	25.33%	26.9%	18.7%
% Other	7.3%	5.8%	10.36%	11.1%	7.0%
Count	46,431,132	30,943,694	15,487,438	12,575,694	2,911,744

Panel B: Loan disposition based on 2007–2015 HMDA					
	All lenders	Traditional banks	Shadow banks	Shadow banks	
				Non-fintech	Fintech
Not sold	23.32%	31.15%	7.50%	6.80%	10.53%
Sold to:					
Fannie Mae	23.37%	23.08%	22.80%	20.25%	33.85%
Freddie Mac	14.63%	17.58%	8.84%	8.23%	11.33%
Ginnie Mae	10.55%	9.12%	13.47%	13.19%	14.66%
Private securitization	0.68%	0.76%	0.49%	0.57%	0.15%
Commercial bank	9.50%	5.38%	17.11%	19.19%	11.28%
Ins(CU)/Mortgage bank	5.93%	2.44%	12.89%	12.34%	15.26%
Affiliate institution	4.75%	6.70%	0.88%	0.99%	0.44%
Other	7.26%	3.09%	15.43%	18.42%	2.49%
Count	46,431,132	30,943,694	15,487,438	12,575,694	2,911,744

Table 2

Time between origination and sale.

Table 2 shows the results of the time-to-sale regression for quarters between origination and sale, using Fannie Mae and Freddie Mac origination data from 2010 to 2015. Columns (1) and (2) compare shadow banks to traditional banks for the entire sample of lenders. Columns (3) and (4) compare present the results with shadow banks broken out by fintech and non-fintech lenders. Columns (5) and (6) compare fintech shadow banks to non-fintech shadow banks among the shadow bank sample only. Columns (1), (3), and (5) have quarter fixed effects and no other controls. Columns (2), (4), and (6) have borrower and loan controls and zip-quarter fixed effects. The left-hand-side variable is in quarters since origination. Its mean among all lenders is 0.46, or approximately 41 days; its mean among shadow bank lenders is 0.40, or approximately 36 days. Standard errors are clustered at the zip-quarter level; t-statistics in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	(1) Qtrs to sale	(2) Qtrs to sale	(3) Qtrs to sale	(4) Qtrs to sale	(5) Qtrs to sale	(6) Qtrs to sale
Sample	All lenders			Shadow banks only		
Shadow bank	−0.103*** (−52.77)	−0.100*** (−52.67)	–	–	–	–
Non-fintech shadow bank	–	–	−0.0812*** (−39.12)	−0.0803*** (−39.37)	–	–
Fintech shadow bank	–	–	−0.189*** (−63.17)	−0.173*** (−60.18)	−0.0846*** (−28.21)	−0.0842*** (−26.84)
Borrower and loan controls	No	Yes	No	Yes	No	Yes
Zip x Quarter FE	No	Yes	No	Yes	No	Yes
Quarter FE	Yes	No	Yes	No	Yes	No
N	4,075,985	40,714,65	4,075,985	4,071,465	1,187,390	1,185,846
R ²	0.0349	0.0491	0.0368	0.0507	0.0603	0.0931

5.3 Tables 3–5: shadow banks serve regulated segments; fintech specializes in refinancing

Table 3

Shadow bank, fintech presence, and the borrower and loan characteristics: all loans.
Table 3 presents descriptive statistics for borrowers and loans by lender type. Panel A summarizes differences in borrower demographics in accepted mortgage applications as reported in the HMDA data. Columns (1)–(4) compare over the period 2007–2015. Columns (5)–(8) cover the 2015. Columns (1) and (2) and (5) and (6) compare traditional and shadow banks; Columns (3) and (4) and (7) and (8) compare non-fintech and fintech shadow banks. Panel B shows the result of β_1 , a linear probability model regressing whether the lender is a shadow bank, Columns (1) and (2); a non-fintech shadow bank, Columns (3) and (4); a fintech lender among all lenders, Columns (5) and (6); or a fintech lender among shadow banks, Columns (7) and (8), on borrower characteristics over the period 2007–2015. Odd columns include year fixed effects. Even columns include year-county fixed effects. For race dummies, the base category is White; for sex dummies, the base is Male; for loan purpose dummies, the base is Purchase; for purchaser dummies, the base is Not sold. For type dummies, the base is Conventional. Coefficients are in percent. Standard errors (in parentheses) are clustered at the county-year level; t -statistics in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Panel A: summary statistics based on (HMDA)							
	2007–2015		2015				
	Traditional banks	Shadow banks	Traditional banks	Shadow banks	Traditional banks	Shadow banks	
		Non-fintech	Fintech		Non-fintech	Fintech	
Count	30,943,694	15,487,438	12,575,694	2,911,744	2,300,721	1,670,680	511,974
Median income	\$83,000	\$79,000	\$78,000	\$82,000	\$89,000	\$79,000	\$82,000
Male	66.98%	67.61%	68.94%	61.87%	65.64%	65.94%	69.01%
Race							
Native American	0.52%	0.50%	0.50%	0.50%	0.57%	0.56%	0.62%
Asian	5.21%	5.79%	6.09%	4.50%	5.63%	5.79%	4.55%
African American	4.72%	5.59%	5.85%	4.48%	4.83%	6.80%	4.80%
Native Hawaiian	0.36%	0.42%	0.43%	0.34%	0.36%	0.45%	0.33%
White	78.04%	76.47%	77.76%	70.90%	77.10%	74.93%	78.17%
Other/Unknown	11.15%	11.23%	9.36%	19.27%	11.49%	12.19%	8.39%
Loan purpose							
Home improvement	6.34%	0.68%	0.78%	0.22%	9.80%	1.14%	1.35%
Refinancing	60.40%	52.69%	47.26%	76.13%	44.67%	47.44%	40.87%
							68.88%

(continued on next page)

Table 3
(continued)

Sample	Panel B: regressions (HMDA)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Shadow bank	Shadow bank	Non-fintech	Non-fintech	Fintech	Fintech	Fintech	Fintech
All lenders								
Shadow banks only								
Income (000 \$)	-0.00857*** (0.0000462)	-0.00673*** (0.0000453)	-0.00602*** (0.0000440)	-0.00468*** (0.0000431)	-0.00165*** (0.0000266)	-0.00206*** (0.0000266)	0.00138*** (0.0000107)	-0.00197*** (0.0000106)
Loan amount (000 \$)	0.00835*** (0.0000392)	0.000881*** (0.0000423)	0.00931*** (0.0000373)	0.000910*** (0.0000402)	-0.000965*** (0.0000238)	-0.0000288 (0.0000249)	-0.0152*** (0.0000799)	-0.00511*** (0.0000895)
Race (Omitted category = White)								
Native American	-0.227*** (0.0877)	-1.362*** (0.0861)	-0.821*** (0.0835)	-2.028*** (0.0820)	0.593*** (0.0506)	0.666*** (0.0506)	1.170*** (0.138)	1.879*** (0.136)
Asian	3.923*** (0.0276)	1.294*** (0.0283)	4.384*** (0.0263)	1.498*** (0.0269)	-0.461*** (0.0159)	-0.286*** (0.0166)	-3.313*** (0.0414)	-2.288*** (0.0421)
Black	0.405*** (0.0190)	0.296*** (0.0194)	0.338*** (0.0285)	0.346*** (0.0296)	0.0676*** (0.0173)	-0.0501*** (0.0179)	0.000883 (0.0448)	0.171*** (0.0456)
Hawaiian	1.363*** (0.103)	-0.696*** (0.101)	1.630*** (0.0980)	-0.774*** (0.0961)	-0.267*** (0.0594)	0.0776 (0.0593)	-0.798*** (0.153)	0.680*** (0.151)
Unknown	7.438*** (0.0292)	5.663*** (0.0286)	3.746*** (0.0278)	1.958*** (0.0272)	3.723*** (0.0168)	3.705*** (0.0168)	5.309*** (0.0407)	5.925*** (0.0400)
NA	-24.98*** (0.816)	-20.46*** (0.794)	-26.68*** (0.776)	-22.71*** (0.756)	1.706*** (0.470)	2.247*** (0.467)	-6.432*** (1.403)	-4.702*** (1.429)
Sex (Omitted category = Male)								
Female	0.163*** (0.0146)	-0.113*** (0.0143)	-0.204*** (0.0139)	-0.535*** (0.0136)	0.366*** (0.0084)	0.422*** (0.00838)	0.896*** (0.0225)	1.110*** (0.0220)
Unknown	-4.636*** (0.0375)	-4.091*** (0.0366)	-8.296*** (0.0357)	-7.846*** (0.0348)	3.660*** (0.0216)	3.755*** (0.0215)	15.78*** (0.0556)	15.33*** (0.0545)
NA	16.27*** (0.734)	16.38*** (0.715)	18.76*** (0.699)	3.487*** (0.681)	-3.862*** (0.424)	-3.822*** (0.420)	-8.503*** (0.709)	-8.503*** (0.692)
Purpose (Omitted category = Purchase)								
Home improvement	-13.26*** (0.0327)	-12.18*** (0.0323)	-11.40*** (0.0311)	-12.25*** (0.0307)	-1.011*** (0.0189)	-0.722*** (0.0190)	-5.862*** (0.116)	-5.399*** (0.114)
Refinance	-2.056*** (0.0148)	-1.792*** (0.0141)	-7.965*** (0.0136)	-7.965*** (0.0134)	5.908*** (0.00823)	6.047*** (0.00831)	18.24*** (0.0212)	18.14*** (0.0213)
Purchaser (Omitted category = Held)								
Fannie Mae	20.65*** (0.0188)	19.04*** (0.0187)	15.66*** (0.0179)	13.96*** (0.0178)	4.995*** (0.0109)	5.082*** (0.0109)	-5.948*** (0.0414)	-4.915*** (0.0415)
Ginnie Mae	19.03*** (0.0333)	12.96*** (0.0326)	12.51*** (0.0316)	6.504*** (0.0310)	6.529*** (0.0192)	6.529*** (0.0192)	-5.494*** (0.0515)	-5.567*** (0.0512)
Freddie Mac	8.171*** (0.0213)	7.333*** (0.0210)	7.620*** (0.0202)	6.654*** (0.0198)	0.551*** (0.0122)	0.679*** (0.0123)	-8.832*** (0.0485)	-8.892*** (0.0484)
Farmer Mac	64.94*** (1.094)	59.91*** (1.067)	65.83*** (1.041)	59.75*** (1.015)	-0.903 (0.631)	0.159 (0.627)	-21.07*** (1.097)	-16.13*** (1.075)
Private securitization	9.693*** (0.0751)	8.028*** (0.0735)	11.82*** (0.0715)	8.11*** (0.0700)	-2.123*** (0.0433)	-2.081*** (0.0432)	-16.46*** (0.142)	-15.03*** (0.142)
Bank	48.21*** (0.0255)	46.72*** (0.0253)	42.39*** (0.0243)	40.79*** (0.0240)	5.824*** (0.0147)	5.923*** (0.0147)	-13.54*** (0.0429)	-13.08*** (0.0429)
Insr or Fnce co.	57.96*** (0.0298)	56.34*** (0.0294)	43.68*** (0.0284)	41.76*** (0.0280)	14.29*** (0.0172)	14.29*** (0.0173)	-3.679*** (0.0444)	-2.688*** (0.0448)
Affiliate	-2.121*** (0.0327)	-3.604*** (0.0325)	-1.137*** (0.0311)	-1.520*** (0.0309)	-0.983*** (0.0189)	-1.084*** (0.0191)	-13.48*** (0.107)	-12.39*** (0.107)
Other	58.01*** (0.0279)	55.87*** (0.0276)	57.20*** (0.0265)	54.89*** (0.0263)	0.807*** (0.0161)	0.972*** (0.0163)	-20.90*** (0.0432)	-20.12*** (0.0439)
Loan Type (Omitted category = Conventional)								
FHA	9.418*** (0.0249)	9.242*** (0.0245)	9.085*** (0.0237)	8.938*** (0.0231)	0.332*** (0.0143)	0.304*** (0.0144)	-1.755*** (0.0301)	-1.972*** (0.0299)
VA	2.331*** (0.0361)	3.389*** (0.0360)	2.717*** (0.0344)	3.237*** (0.0343)	-0.386*** (0.0209)	-0.0475 (0.0212)	-1.909*** (0.0461)	-1.713*** (0.0464)
FSA/RHS	-1.319*** (0.0566)	-0.660*** (0.0563)	-2.205*** (0.0538)	2.484*** (0.0536)	-2.832*** (0.0326)	-3.144*** (0.0331)	-5.681*** (0.0790)	-6.684*** (0.0765)
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Year x County FE	No	Yes	No	Yes	No	Yes	No	Yes
N	43,138,392	43,138,392	43,138,392	43,138,392	43,138,392	43,138,392	14,340,698	14,340,698
R ²	0.241	0.281	0.223	0.265	0.059	0.074	0.133	0.178

Read: Table 3 shows that shadow-bank borrowers have about \$4,000 lower income than bank borrowers over 2007–2015, with the gap widening to about \$9,000 by 2015. Within markets, lower-income borrowers and racial minorities are more likely to use shadow banks. FHA loans are about 9 percentage points more likely to be originated by shadow banks. Tables 3 and 4 also show a very strong refinance tilt for fintech: among shadow-bank loans, a refinance is nearly 20 percentage points more likely to be a fintech loan in HMDA, and about 7 to 8 percentage points more likely in the conforming-loan data. Table 5 shows that counties with more unemployment, more minority residents, and higher FHA shares have more shadow-bank penetration, while counties where more residents have stayed in the same house for at least a year have more fintech penetration.

Economic message. The shadow-bank facts line up with a regulatory-arbitrage story: nonbanks grow where bank lending to riskier or more scrutinized borrowers becomes relatively unattractive. The fintech facts line up with a technology/convenience story: online origination is especially good at refinancing standardized loans for borrowers already in homes.

5.4 Tables 6 and 7: fintech charges a premium and loans prepay more

Read: Table 6 shows essentially no large average rate difference between all shadow banks and traditional banks once controls are added: the coefficient is only about 0.007 percentage points, less than 1 basis point. But composition matters a lot. Non-fintech shadow banks charge about 2.4 basis points less than banks, while fintech shadow banks charge about 12.9 basis points more

Table 4

Shadow bank presence and the borrower and loan characteristics: conforming loans.

Table 4 shows the results of a linear probability model, Eq. (1), regressing whether the lender is a shadow bank, Columns (1) and (2); a non-fintech shadow bank, Columns (3) and (4); a fintech lender among all lenders, Columns (5) and (6); or a fintech lender among shadow banks, Columns (7) and (8), on individual characteristics, using the pooled Fannie Mae and Freddie Mac Data for the period 2010–2015. Odd columns include quarter fixed effects only; even columns include zip-quarter fixed effects. Loan purpose dummies (Refinance, Investment/Second home) use Purchase and Primary residence as the base category. Coefficients are in percents. Standard errors are clustered by zip-quarter; *t*-statistics in parentheses; * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Shadow bank	Shadow bank	Non-fintech	Non-fintech	Fintech	Fintech	Fintech	Fintech
Sample	All lenders				Shadow banks only			
Loan amount	0.0000100*** (90.27)	0.00000748*** (55.75)	0.0000103*** (99.88)	0.00000846*** (67.29)	-0.00000292*** (-5.04)	-0.000000975*** (-14.19)	-0.0000122*** (-52.91)	-0.0000104*** (-37.45)
Loan term (Months)	0.0436*** (212.38)	-0.00237*** (-10.78)	0.0231*** (168.56)	-0.0000162 (-0.80)	0.0115*** (116.40)	-0.00221*** (-20.48)	0.00513*** (10.18)	-0.0176*** (-30.33)
Loan-to-value	-0.0608*** (-58.29)	-0.0414*** (-38.33)	-0.0400*** (-41.31)	-0.0295*** (-29.35)	-0.0208*** (-39.71)	-0.0119*** (-21.43)	-0.0392*** (-17.58)	-0.0293*** (-12.62)
Debt-to-income	0.0606*** (41.09)	0.0517*** (35.71)	0.0208*** (19.48)	0.0223*** (16.60)	0.0338*** (45.73)	0.0292*** (39.44)	0.302*** (32.92)	0.0789*** (25.84)
FICO	-0.0186*** (-51.68)	-0.0207*** (-59.23)	0.000627 (1.90)	-0.0192*** (-5.15)	-0.0191*** (-49.90)	-0.0519*** (-94.54)	-0.0416*** (-71.09)	-0.0416*** (-58.69)
Investment/Secondary property	-1.333*** (-30.50)	-2.431*** (-54.68)	-0.0902* (-2.22)	-0.772*** (-18.63)	-1.243*** (-31.63)	-1.659*** (-40.37)	-3.359*** (-83.47)	-4.226*** (-105.76)
Refinance	-0.623*** (-21.90)	1.501*** (51.08)	-2.484*** (-83.23)	-0.822*** (-30.39)	1.862*** (133.10)	8.083*** (339.77)	6.832*** (178.63)	6.832*** (178.63)
First-time buyer	-5.805*** (-139.23)	-4.725*** (-115.60)	-1.808*** (-45.56)	-0.847*** (-21.40)	-3.997*** (-113.31)	-3.878*** (-106.91)	-11.79*** (-348.39)	-11.79*** (-348.39)
Has mortgage insurance	1.282*** (27.64)	0.909*** (19.91)	0.597*** (13.83)	0.349*** (8.27)	0.685*** (27.47)	0.551*** (22.19)	2.637*** (28.96)	2.023*** (22.73)
Zip x Quarter FE	No	Yes	No	Yes	No	Yes	No	Yes
Quarter FE	Yes	No	Yes	No	Yes	No	Yes	No
N	8480852	8480851	8480852	8480851	8480852	8480851	1946017	1946017
R ²	0.0709	0.1315	0.0404	0.101	0.0364	0.0736	0.0581	0.162

Table 6

Shadow bank and fintech mortgage rates: conforming loans.

Table 6 shows the results of Eq. (7) using Fannie Mae and Freddie Mac loans from 2010–2015. Columns (1) and (2) test differences between shadow banks and traditional banks. Columns (3) and (4) split shadow banks into fintech and non-fintech lenders and compare interest rates across all lenders. Columns (5) and (6) test differences in fintech rates within shadow banks. Columns (1), (3), and (5) quarter fixed effects and no other controls. Columns (2), (4), (6) have quarter times zip fixed effects and borrower controls. Standard errors are clustered at the zip-quarter level. Interest rates are quoted in percent. The mean interest rate over the sample period is 4.74. *t*-statistics in parentheses; * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001.

	(1)	(2)	(3)	(4)	(5)	(6)
	Interest rate	Interest rate	Interest rate	Interest rate	Interest rate	Interest rate
Sample	All lenders			Shadow banks only		
Shadow bank	0.00665*** (5.19)	0.00714*** (8.33)	-	-	-	-
Non-fintech shadow bank	-	-	-0.0281*** (-20.48)	-0.0242*** (-27.42)	-	-
Fintech shadow bank	-	-	0.143*** (87.68)	0.129*** (101.99)	0.163*** (91.09)	0.144*** (113.17)
Borrower and loan controls	No	Yes	No	Yes	No	Yes
Zip x Quarter FE	No	Yes	No	Yes	No	Yes
Quarter FE	Yes	No	Yes	No	Yes	No
N	8,485,573	8,480,376	8,485,573	8,480,376	1,946,802	1,943,693
R ²	0.598	0.808	0.601	0.811	0.585	0.807

Table 5

Shadow bank and fintech penetration and regional characteristics.

Table 5 shows the regional characteristics associated with shadow bank and fintech penetration. Panel A summarizes demographic differences between counties with low and high shares of shadow bank lending in 2015. Shadow bank and fintech share is calculated from accepted HMDA acceptances. Demographic information comes from the American Community Survey, while Herfindahl, Numbers of lenders, and Percentage of FHA loans is calculated from HMDA. Column (1) shows statistics for all counties. Column (2) shows statistics for counties in the bottom 25% of shadow bank share. Column (3) shows statistics for counties in the top 25% of shadow bank share. Column (4) shows statistics for counties in the bottom 25% of fintech share. Column (5) shows statistics for counties in the top 25% of fintech share. Panel B shows the results of Eq. (4), where the share of shadow banks—Columns (1)–(3)—or fintech—Columns (4)–(6)—in a county is regressed on county characteristics. Columns (1) and (4) are the baseline specification. Columns (2) and (5) include the county-level Herfindahl measure. Columns (3) and (6) include the number of unique lenders within a county. *t*-statistics in parentheses; * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001.

Panel A: summary statistics						
Median values	All counties	Shadow bank		Fintech		
		Bottom quartile	Top quartile	Bottom quartile	Top quartile	
Median household income	\$45,114.00	\$44,587.00	\$46,949.00	\$48,160.00	\$41,101.00	
Population density	42.7	35.6	44.1	55.3	19.1	
% with less than 12th grade education	13.10%	11.80%	15.35%	10.80%	17.00%	
% with bachelor degree or higher	18.20%	17.70%	18.20%	20.00%	15.40%	
% African American	2.10%	1.06%	2.83%	1.35%	1.81%	
% Hispanic	3.76%	2.40%	8.80%	3.39%	4.07%	
Unemployment rate	7.00%	6.40%	7.50%	6.30%	7.20%	
% living in same house ≥ 1 year	86.50%	87.60%	86.1%	86.74%	87.25%	
Herfindahl	0.0976%	0.1484%	0.0783%	0.1156%	0.1142%	
# Lenders	39.00	29.00	50.00	45.00	22.00	
% of FHA origination loans	16.28%	12.50%	18.71%	13.58%	17.76%	
Population	259,930.00	209,130.00	314,700.00	348,840.00	134,720.00	
% with less than 35 K salary	26.70%	27.10%	25.70%	24.80%	29.95%	
Panel B: regressions						
	(1)	(2)	(3)	(4)	(5)	(6)
	% Shadow banks	% Shadow banks	% Shadow banks	% Fintech	% Fintech	% Fintech
Median household income	0.000199*** (0.0000224)	0.000139*** (0.00000221)	0.000135*** (0.00000231)	-0.0000368*** (0.0000094)	-0.0000309*** (0.00000951)	0.00000211 (0.00000955)
Population density	-0.000620*** (0.000146)	-0.000606*** (0.000143)	-0.000665*** (0.000144)	-0.000256*** (0.0000616)	-0.000258*** (0.0000614)	-0.000229*** (0.0000597)
% with less than 12th grade education	0.185*** (0.0469)	0.147*** (0.0458)	0.152*** (0.0462)	0.0817*** (0.0197)	0.0764*** (0.0197)	0.0780*** (0.0191)
% with bachelor degree or higher	0.0743*** (0.0331)	0.0439*** (0.0331)	-0.0263*** (0.0331)	-0.00912*** (0.0143)	-0.00493*** (0.0142)	0.0522*** (0.0145)
% African American	0.0511*** (0.0151)	0.0414*** (0.0147)	0.0376*** (0.0148)	0.0219*** (0.0634)	0.0233*** (0.0631)	0.0302*** (0.0617)
% Hispanic	0.0170*** (0.0170)	0.0259*** (0.0166)	0.0238*** (0.0169)	0.0860*** (0.00714)	0.0846*** (0.00713)	0.0979*** (0.00698)
Unemployment rate	0.450*** (0.0631)	0.321*** (0.0623)	0.252*** (0.0654)	-0.119*** (0.0265)	-0.100*** (0.0268)	0.0045 (0.0271)
% living in same house ≥ 1 year	-0.100*** (0.0444)	-0.0493*** (0.0415)	-0.0045*** (0.0440)	0.114*** (0.0187)	0.106*** (0.0187)	0.0872*** (0.0182)
% FHA	0.288*** (0.0213)	0.259*** (0.0209)	0.271*** (0.0210)	0.0744*** (0.00894)	0.0788*** (0.00897)	0.0848*** (0.00870)
Herfindahl	-	-19.81*** (1.549)	-	-	-2.822*** (0.666)	-
# Lenders	-	-	0.0584*** (0.00605)	-	-	-0.0355*** (0.00250)
Constant	12.15*** (4.124)	14.83*** (4.026)	12.47*** (4.065)	-2.647*** (1.735)	-3.042*** (1.732)	-2.845*** (1.682)
N	3131	3131	3131	3131	3131	3131
R ²	0.256	0.293	0.277	0.176	0.181	0.226

Table 7

Shadow bank presence and loan performance: conforming loans.

Table 7 shows the results of Eq. (8) for loan outcomes using Fannie Mae and Freddie Mac performance data from 2010 to 2013. Panel A shows the results for default. Panel B shows the results for prepayment. Prepayment is defined as the loan being prepaid within two years of origination. Default is defined as the loan status becoming 60 days past due within two years of origination. Columns (1)–(2) test differences between shadow banks and traditional banks. Columns (3)–(4) split shadow banks into fintech and non-fintech lenders and compare performance across all lenders. Columns (5)–(6) test differences in fintech performance within shadow banks. Columns (1), (3), and (5) quarter fixed effects and no other controls. Columns (2), (4), (6) have quarter times zip fixed effects and borrower controls. The left-hand-side variable is in percent. Its mean for defaults over the sample period is 0.23. Its mean for prepayments over the sample period is 11. Standard errors are clustered at the zip-quarter level; *t*-statistics in parentheses; * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001.

Panel A: Default						
	(1)	(2)	(3)	(4)	(5)	(6)
	Defaulted	Defaulted	Defaulted	Defaulted	Defaulted	Defaulted
Sample	All lenders			Shadow banks only		
Shadow bank	0.0196*** (3.90)	0.0208*** (4.11)	-	-	-	-
Non-fintech shadow bank	-	-	0.0116* (2.15)	0.0236*** (4.34)	-	-
Fintech shadow bank	-	-	0.0557*** (4.66)	0.00795 (0.67)	0.0307* (2.33)	-0.0286* (-2.04)
Borrower and loan controls	No	Yes	No	Yes	No	Yes
Zip x Quarter FE	No	Yes	No	Yes	No	Yes
Quarter FE	Yes	No	Yes	No	Yes	No
N	6,527,612	6,523,402	6,527,612	6,523,402	1,151,439	1,149,115
R ²	0.000359	0.0112	0.000362	0.0112	0.000609	0.0348
Panel B: Prepayment						
	(1)	(2)	(3)	(4)	(5)	(6)
	Prepaid	Prepaid	Prepaid	Prepaid	Prepaid	Prepaid
Sample	All lenders			Shadow banks only		
Shadow bank	2.469*** (20.31)	1.823*** (23.38)	-	-	-	-
Non-fintech shadow bank	-	-	1.456*** (10.71)	0.713*** (8.94)	-	-
Fintech shadow bank	-	-	7.054*** (34.50)	6.757*** (34.23)	5.675*** (26.48)	6.358*** (30.27)
Borrower and loan controls	No	Yes	No	Yes	No	Yes
Zip x Quarter FE	No	Yes	No	Yes	No	Yes
Quarter FE	Yes	No	Yes	No	Yes	No
N	6,527,612	6,523,402	6,527,612	6,523,402	1,151,439	1,149,115
R ²	0.0566	0.151	0.0571	0.152	0.0594	0.155

than banks and about 14.4 basis points more than non-fintech shadow banks. Table 7 shows only slightly higher default for shadow-bank loans — about 0.02 percentage points over two years — but much higher prepayment. Shadow-bank loans are about 1.8 to 2.5 percentage points more likely to prepay, and fintech loans are roughly 6.8 to 7.1 percentage points more likely to prepay than bank loans.

Economic message. The pricing evidence rejects the simple “fintech is cheaper” story. Fintech wins while charging more, not less, which is consistent with differentiated convenience or superior matching. The performance results also suggest that shadow-bank and especially fintech borrowers are more refinance-sensitive, which fits the specialization result.

5.5 Table 8 and Figure 6: regulation predicts where shadow banks expand

Table 8

Regulatory activity and shadow bank market shares.

Table 8 shows the result of the regressions specified by Eqs. (16) and (17). The regression is at the county level. Panel A, Columns (1)–(4) regress percent changes in county-level shadow bank market share on four measures of regulatory exposure. Columns (5)–(8) regress percent changes in county-level shadow bank lending volume on four measures of regulatory exposure. All columns include demographic controls. Δ Capital ratio is the 2008 lending-weighted change in bank capital ratios between 2008 and 2015. % MSR is the lending-weighted 2008 fraction of MSR assets of tier-1 capital. Lawsuit exposure is the lending-weighted exposure to mortgage-related lawsuits. % OTS is the share of mortgage originations regulated by the OTS in 2008. Panel B shows the results of using 2006–2008 capital ratio changes to instrument for 2008–2015 changes. Column (1) is the first stage; Columns (2) and (4) are the reduced form regressions. Columns (3) and (5) are the second stage regressions. The F -statistic of the instrument is 48.68. All regulatory variables are standardized, so coefficients can be interpreted as percent change in shadow bank share for a one standard deviation change in the regulatory measure. t -statistics are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Panel A: regulation and market share changes								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Δ Shadow bank share				Δ Shadow bank lending			
Δ Capital ratio	3.358*** (16.601)	–	–	–	3.412*** (8.160)	–	–	–
% MSR	–	1.902*** (9.403)	–	–	–	1.939*** (4.739)	–	–
Lawsuit exposure	–	–	2.034*** (10.898)	–	–	–	6.494*** (17.921)	–
% OTS	–	–	–	1.215*** (4.759)	–	–	–	5.209*** (10.333)
County controls	Y	Y	Y	Y	Y	Y	Y	Y
N	3094	3094	3094	3094	3094	3094	3094	3094
R ²	0.211	0.164	0.172	0.147	0.327	0.318	0.378	0.336

Panel B: 2006–2008 capital ratio change as instrument					
	(1)	(2)	(3)	(4)	(5)
	Δ CR (2008–2015) first stage		Δ Shadow bank share		Δ Shadow bank lending
Δ Capital ratio (2008–2015) (Fit)	–	–	9.634*** (8.270)	–	8.241*** (3.838)
Δ Capital ratio (2006–2008)	–	–0.238*** (–11.270)	–2.289*** (–9.201)	–	–1.958*** (–3.888)
County controls	Y	Y	Y	Y	Y
N	3094	3094	3094	3094	3094
R ²	0.215	0.163	–0.036	0.316	0.298

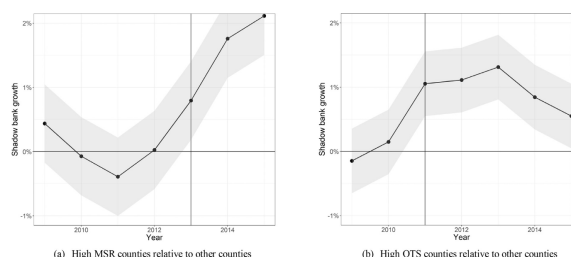


Fig. 6. Mortgage servicing rights, OTS shares, and bank shares over time. Fig. 6 shows the year-by-year relationship in the county-level shadow bank market share for counties in the top quintile of MSR assets (a) or the top quintile of OTS originations (b). Shadow bank market shares in high MSR and high OTS counties are compared with other counties. Coefficients are estimated from the difference in difference regression specified in Eq. (16). $\text{ShadowBankShare}_{it} = \gamma_0 + \gamma_1 + \beta (\text{Year} \times \text{High}) + (\text{Year} \times \text{X}_{it}) + \epsilon_{it}$. The shaded regions denote 95% confidence intervals. The rule implementing the MSR guidelines was finalized in 2013. The OTS dissolved in 2011.

Read: Table 8 is the paper’s central regulation evidence. A one-standard-deviation increase in local bank capital-ratio rebuilding is associated with about a 3.4 percentage point rise in shadow-bank market share and a 3.4 percentage point rise in shadow-bank lending growth. A one-standard-deviation increase in MSR exposure is associated with about 1.9 percentage points more shadow-bank market-share growth and 1.9 percentage points more shadow-bank lending growth. Lawsuit exposure predicts about 2.0 percentage points more shadow-bank market share and 6.5 percentage points more shadow-bank lending growth. OTS exposure predicts about 1.2 percentage points more market share and 5.2 percentage points more lending growth. Figure 6 then shows that the shadow-bank-share gap opens precisely around the MSR rule finalization in 2013 and the OTS closure in 2011.

Economic message. This is the paper’s best evidence for regulatory arbitrage. Shadow banks do not simply grow everywhere. They grow especially where banks are locally hit harder by specific post-crisis regulatory changes.

5.6 Tables 9 and 10 and Figure 7: fintech uses different information and sells convenience

Table 9

R^2 of different specifications explaining interest rates.

Table 9 shows the R^2 of observations for different specifications of the regression specified by Eq. (10). Data is from Fannie Mae and Freddie Mac. Fixed effects are different set as that their effects are not included in the total rank of squares. The table shows pooled regressions between 2010 and 2015 for the bank shadow bank, non-fintech, and fintech subsamples. Nonfintech controls include third-order polynomials of all observables. Tests of significance of R^2 differences follow *Donaldson and Whited (2015)*. *** denotes significance at the 1% level.

Panel A: R^2 of pooled regressions, 2010-2015										
Specifications				Full sample			Shadow bank sample			
Controls	Quarter FE	Zip-Quarter FE	Lender FE	Rank	Shadow bank	Non-Fintech	Fintech	Non-fintech - Fintech	(Non-fintech - Fintech)	
FICO, LTV	Y	N	N	0.155	0.234	0.240	0.059	0.000***	0.000***	
FICO, LTV	N	Y	N	0.0488	0.161	0.189	0.007	0.003***	0.003***	
All	N	N	N	0.047	0.028	0.086	0.008	0.000***	0.000***	
All	Y	N	N	0.067	0.070	0.100	0.045	0.000***	0.000***	
Nonlinear	N	N	N	0.067	0.069	0.100	0.045	0.000***	0.000***	
Nonlinear	N	Y	N	0.033	0.031	0.044	0.013	0.001***	0.001***	
Nonlinear	N	Y	Y	0.039	0.033	0.042	0.030	0.002***	0.002***	

Table 10

Fintech cost and convenience

Table 10 shows the results of the regression specified by Eq. (20). Data are from Fannie Mae and Freddie Mac Shadow Bank originations between 2010 and 2015. High FICO is a dummy variable for borrowers with FICO in the top decile for the year. Columns (1) and (2) show the results for the full sample, 2010-2015. Columns (3) and (4) show the results for the early period, 2010-2013. Columns (5) and (6) show the results for the late sample, 2014-2015. All columns include borrower and loan controls. Columns (1), (3), and (5) include quarter fixed effects; Columns (2), (4), and (6) include quarter-sip fixed effects. The left-hand side variable is in percent term; the mean is 4.38. Standard errors are clustered at the zip-quarter level. t -statistics are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Full (2010-2015)		Early (2010-2013)		Late (2014-2015)	
	(1) Rate	(2) Rate	(3) Rate	(4) Rate	(5) Rate	(6) Rate
Fintech	0.156*** (121.33)	0.143*** (112.12)	0.140*** (77.41)	0.122*** (70.28)	0.192*** (99.07)	0.169*** (97.99)
High FICO $\times$ Fintech	0.00954*** (3.50)	0.00338* (2.46)	0.00606*** (4.08)	0.00770*** (3.53)	0.0111*** (5.32)	0.00948*** (4.55)
Borrower and loan controls	Yes	Yes	Yes	Yes	Yes	Yes
Zip $\times$ Quarter FE	No	Yes	No	Yes	No	Yes
Quarter FE	Yes	No	Yes	No	Yes	No
N	1,946,017	1,943,093	1,151,009	1,148,115	795,008	794,978
R^2	0.792	0.808	0.826	0.841	0.882	0.898

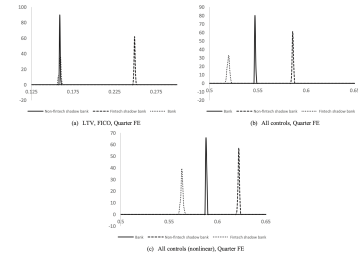


Fig. 7. Distribution of bootstrapped R^2 . Fig. 7 shows the distribution of bootstrapped R^2 , corresponding to the determinants of interest rates. Each bootstrapped sample reflects a random sample of originations with replacement, across the interest rate regressors, and computes the R^2 . The bootstrapping is run on 100 random samples. Panel A shows a model of interest rates with FICO, LTV, and quarter fixed effects. Panel B shows a model of interest rates with all (linear) observables and quarter fixed effects. Panel C shows a model of all observables with zip or zip-quarter errors included.

Read: Table 9 shows that FICO and LTV explain nearly 25% of the variation in non-fintech shadow-bank rates but less than 16% of the variation in fintech rates. Even in more saturated specifications, the R^2 stays above 54% for non-fintech and below 52% for fintech. Figure 7 visualizes these differences. Table 10 then shows that fintech borrowers pay a premium: roughly 14.3 to 15.6 basis points in the full sample, with an additional premium of about 0.3 to 0.6 basis points for top-decile FICO borrowers and about 1.0 to 1.1 basis points in the late sample.

Economic message. Two separate technology facts emerge. First, fintech lenders appear to price with a different information set. Second, borrowers — especially high-FICO borrowers — are willing to pay more for fintech, which supports the convenience/quality interpretation.

5.7 Figures 8 and 9: the model says regulation matters most for total shadow-bank growth

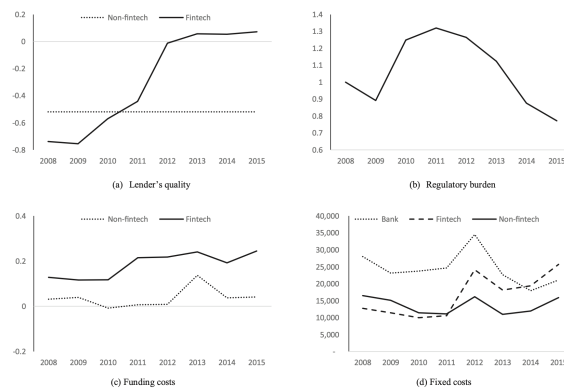


Fig. 8. Calibrated characteristics of lenders. Fig. 8 presents the model parameters discussed in Section 9.3. Panel A shows lender quality characteristics for fintech and non-fintech shadow banks relative to traditional bank. Panel B shows the evolution of regulatory burden face by traditional banks implied by our model relative to 2008 level. A higher value of the parameter implies a lower regulatory burden level. Panel C shows funding costs for fintech and non-fintech shadow banks relative to traditional bank. Panel D shows fixed costs of traditional banks, fintech and non-fintech shadow banks.

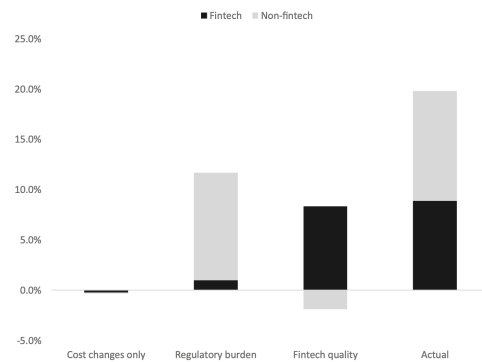


Fig. 9. Counterfactuals for the change in the shadow bank market share implied by our model. Fig. 9 shows predicted changes in shadow bank market share in the overall mortgage market between 2008 and 2015, broken down between non-fintech and non-fintech entrants, for three counterfactuals regarding fintech quality and bank regulatory impairment. “No changes” fixes both fintech quality to its 2008 and bank regulatory burden parameter to zero. “Regulatory impairment” has fixed fintech quality and allows bank regulatory burden to vary as calibrated. “Fintech quality increase” fixes bank regulatory burden and allows fintech quality to vary as in the data. “Actual” shows the actual changes in our data.

Read: Figure 8 shows the calibrated wedges over time: banks’ regulatory burden worsens sharply after 2011, while fintech quality improves relative to non-fintech. Figure 9 reports counterfactual decompositions. If neither regulatory burden nor fintech quality improved relative to 2008, shadow-bank growth is near zero. Allowing only bank regulatory burden to worsen generates about 12 percentage points of shadow-bank growth. Allowing only fintech quality to improve generates about 6 percentage points of shadow-bank growth, with fintech gaining roughly 8.3 percentage points and non-fintech losing roughly 2 percentage points.

Economic message. The calibrated model translates the reduced-form findings into a simple quantitative ranking: regulation explains most of total shadow-bank growth, while technology explains most of fintech growth specifically.

6 Short summary

- Shadow banks' mortgage-origination share rises from about 30% in 2007 to 50% in 2015, and their FHA share rises to about 75%.
- Fintech lenders rise from about 3% to 12% of all mortgages and become an important component of shadow-bank growth.
- Shadow banks rely heavily on originate-to-distribute funding and GSE/FHA channels, while banks keep far more loans on balance sheet.
- Shadow banks are especially active in segments where banks face greater post-crisis regulatory scrutiny: FHA, lower-income, minority, and legally risky markets.
- Fintech lenders specialize in refinancing, use pricing models less tied to standard hard information, and sell loans faster after origination.
- Fintech conforming mortgages are not cheaper. Relative to non-fintech shadow banks, fintech charges roughly 14–16 basis points more on average.
- High-FICO borrowers pay an even larger fintech premium, consistent with willingness to pay for convenience.
- Counties more exposed to bank capital rebuilding, MSR treatment, lawsuits, and OTS closure see larger increases in shadow-bank market share and lending.
- The calibrated model attributes about 60% of total shadow-bank growth to regulation and about 30% to technology; technology explains about 90% of fintech growth.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the post-crisis rise of shadow banks in mortgage origination is not just a broad secular trend in “nonbanks.” It is closely linked to regulatory pressure on banks. At the same time, the rise of fintech within shadow banking reflects more than simple cost cutting: fintech appears to offer a differentiated product that borrowers value and pay for.

7.2 Economic intuition

Banks entered the post-crisis period with damaged balance sheets and then faced tighter capital rules, harsher treatment of mortgage servicing rights, more legal exposure, and stricter supervision. These changes made some mortgage activity relatively less attractive inside the banking sector. Shadow banks stepped into the gap, using GSE and FHA funding channels rather than deposit

funding. Within the shadow-bank sector, fintech lenders gained additional share because online origination improved convenience and allowed them to price loans with a different information set.

7.3 Takeaway

The paper shows that post-crisis mortgage intermediation shifted along *both* a regulatory margin and a technology margin. Regulation explains why origination left banks; technology explains why some of it went specifically to fintech. The result is a mortgage market that looks less bank-centered but still deeply tied to public credit support through the GSEs and FHA.